One Newton Polygon, Two Frobenius Polynomials: A Disproof of Conjecture 00000005556

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Abstract

Over $\mathbb{F}_3$, take the rank-one unit-root F-crystals with Frobenius $+1$ and -1 on the free $\mathbb{Z}_3$-line. Their rationalizations are actual linear automorphisms of $\mathbb{Q}_3$. Both Newton polygons are the horizontal segment from $(0,0)$ to $(1,0)$, and both crystals are pure of weight zero. Their Frobenius characteristic polynomials are nevertheless $X-1$ and $X+1$. Thus the Newton polygon does not determine the characteristic polynomial, even at fixed rank and pure weight in the unit-root case. Lean verifies the integral and rational automorphisms, scalar extension compatibility, valuations, actual matrix characteristic polynomials, geometric Newton segments and the universal non-determination statement.

Reading of the source. The first assertion in both source languages is that the characteristic polynomial is given by the Newton polygon. In particular, this requires equal Newton polygons to give equal characteristic polynomials. We refute this necessary uniqueness condition, which already suffices to refute the stated conjecture.

The final phrase, saying that in the unit-root case something is uniquely determined by pure weight, has an ambiguous subject. If it refers to the characteristic polynomial, the same example also refutes it. If it refers only to the Newton polygon, that weaker claim is not refuted: all rank-one unit-root objects already have the same horizontal Newton polygon. We do not claim otherwise.

The actual crystalline objects. Fix the base point $\operatorname{Spec}\mathbb{F}_3$. Here $W(\mathbb{F}_3)=\mathbb{Z}_3$, its fraction field is $\mathbb{Q}_3$, and the Witt Frobenius is the identity. Thus a free rank-one F-crystal on this point is described by a free $\mathbb{Z}_3$-module M of rank one with an injective linear Frobenius. It is unit-root when Frobenius is an automorphism of this integral lattice. Its rationalization is an automorphism of $M\otimes_{\mathbb{Z}_3}\mathbb{Q}_3$.

This is the specialization of the usual definition with a Frobenius-semilinear automorphism: when the base Frobenius is the identity, semilinearity is ordinary linearity. For the general definition see Jacob Lurie, *Lecture 26–Isocrystals*, Definition 1: lecture notes. The standard identifications with Witt vectors and this specialization are stated here as mathematical background. Lean directly verifies all the resulting integral and rational linear-algebra data over the actual 3-adic rings and field.

Use the following two objects:

	M	F on M and its rationalization
C_+	$\mathbb{Z}_3$	$x \longmapsto x$
C_-	$\mathbb{Z}_3$	$x \longmapsto -x.$

Both maps and their inverses preserve the integral lattice. The integral maps agree, under the inclusion $\mathbb{Z}_3 \hookrightarrow \mathbb{Q}_3$, with the displayed rational maps. There is no quotient, reduction of coefficients, or change of base in this comparison.

Theorem 1. *The two unit-root F-crystals C_+ and C_- have equal Newton polygons and equal pure weight zero but distinct characteristic polynomials. Their rational Frobenius operators are not conjugate by a $\mathbb{Q}_3$-linear isomorphism.*

Proof. The Frobenius matrices in the basis 1 are respectively [1] and [−1], giving

$$\chi_+(X) = X - 1, \qquad \chi_-(X) = X + 1.$$

They differ because $1 \neq -1$ in the characteristic-zero field $\mathbb{Q}_3$.

Let v_3 be the 3-adic valuation with $v_3(3) = 1$. Both eigenvalues are units:

$$|1|_3 = |-1|_3 = 1, \qquad v_3(1) = v_3(-1) = 0.$$

For a monic degree-one polynomial $c_0 + c_1X$ with $c_0 \neq 0$, its Newton polygon is the segment joining $(0, v_3(c_0))$ and $(1, v_3(c_1))$. Thus the polygons of $X - 1$ and $X + 1$ both equal

$$\{(s, 0) : 0 \leq s \leq 1\}.$$

The eigenvalues $1, -1$ are rational algebraic numbers, and every complex embedding of their number field $\mathbb{Q}$ has absolute value one on them. Since $3^{0/2} = 1$, both objects are pure of weight zero in the usual Weil-number sense. This is a stronger counterexample than one obtained by changing rank or weight.

Finally, a linear isomorphism f intertwining the two operators would satisfy $-f(1) = f(1)$, hence $2f(1) = 0$ and $f(1) = 0$. Injectivity would imply $1 = 0$, impossible. □

Why this refutes determination. A Newton polygon records coefficient valuations, or equivalently the corresponding valuation slopes, rather than the full unit parts of Frobenius eigenvalues. The above example preserves every Newton datum while changing the unit $+1$ to -1. Therefore no function of the Newton polygon alone can return the characteristic polynomial for all crystals in the stated class. Even supplying the pure weight zero in addition does not distinguish this pair. This argument stays over the fixed finite base field $\mathbb{F}_3$; it makes no assertion about classification after extension to an algebraically closed base.

Formal verification and scope. The Lean structure `RankOneCrystal` contains an actual $\mathbb{Z}_3$-linear equivalence of the integral line, an actual $\mathbb{Q}_3$-linear equivalence of the rational line, and a proof of scalar extension compatibility. The two examples instantiate all three fields. The theorem

`matrix_represents_frobenius`

proves that the actual one-by-one Mathlib matrix acts as the rational Frobenius on every vector. Characteristic polynomials are Mathlib's `Matrix.charpoly`. The valuations and norms are those of Mathlib's actual 3-adic field, and the Newton polygon is a standard real geometric `segment` joining the two coefficient-valuation points. The valuation calculation does not substitute integers or a hard-coded slope table for 3-adic values.

`PureWeightZero` requires a rational Frobenius eigenvalue and modulus one under every complex embedding of $\mathbb{Q}$; this is exactly the usual purity condition for the rational eigenvalues in our counterexample. It is not intended as a definition of all possibly irrational pure crystals. The principal negations are

`not_newton_determines_characteristic_polynomial`,
`not_pure_weight_determines_characteristic_polynomial`.

These are universally quantified propositions over the actual specialized crystalline objects; the first additionally requires equal Newton polygons and unit-root conditions on both objects. The second treats the stronger characteristic-polynomial reading of the ambiguous final clause. Non-conjugacy is separately proved by `no_frobenius_conjugacy`.

The Witt-vector identification and equivalence of this specialized linear-algebra description with F-crystals on the point are paper background. No formal construction of the full crystalline site or of Mathlib's general Witt-vector isocrystal class is claimed. The finite-field identity $x^3 = x$ on $\mathbb{F}_3$ is also checked directly in Lean.

Source and reproduction. The preserved bilingual source is `SOURCE.md`; the provenance file records the repository commit and source hash. The project pins Lean 4.19.0 and Mathlib revision

`c44e0c8ee63ca166450922a373c7409c5d26b00b`.

From the `lean` directory run `lake build`, then

`lake env lean -DwarningAsError=true Main.lean`.

Principal axiom reports contain only `propext`, `Classical.choice`, and `Quot.sound`. Development and self-review were performed by a delegated AI agent; the coordinating agent reviews the result separately. No external independent review is claimed.